NOVA — Rev B: simulations and system analyses

Date: 2026-10-29 · ngspice 42 (batch) · Python 3 (numpy) · CadQuery mass properties · ronmart · Caril Ronnie Martínez Vera

Everything in Rev B that can be checked without hardware was simulated: power rails, transients at the connectors, signal edges, and the robot as a system (stability, drive, energy, charging, heat, docking). Every model lists its inputs and their source in the scripts (`sim/spice_sims.py`, `sim/system_sims.py`, `sim/mass_props.py`): datasheet values where they exist, CAD for geometry and masses of printed parts, and marked estimates (“est.”) for the rest.

1. Summary

#	Check	Result	Limit	Verdict
S1	VSYS ripple, both motors PWM + Pi 3 A	46 mV p-p driving, 108 mV at stall; min 6.30 V at 6.6 V battery	TPS566238 / TPS62933 inputs ≥ 3 V	OK
S2	+5V (Pi 5) load step 1 → 3 A	min 4.90 V at the Pi, 4.99 V steady at 3 A; ripple 3.9 mV p-p	Pi 5 under-voltage ≈ 4.63 V	OK (270 mV margin)
S2	+5V extreme step 0.5 → 5 A, 2 A/µs, battery 6.6 V	min 4.62 V for ≈ 10 µs; 4.88 V steady; overshoot 5.22 V; L301 peak 6.6 A	4.63 V; Isat 11 A; valley limit ≥ 6.1 A	Marginal (option: 5.22 V)
S3	+3V3 step 0.1 → 0.6 A	3.29 … 3.35 V, ripple 2.1 mV	3.0–3.6 V	OK
S4	Power-up sequence	+3V3 at 5.8 ms, +5V at 4.9 ms, ESP32 EN at 17.4 ms	EN after the supply is stable	OK
S5	Dock: adapter plug-in / robot docks on a live dock / reversed adapter	VAC2 peak 16.5 V / 19.5 V / blocked (0 V)	VAC2 abs. max 30 V	OK
S6	USB-C hot plug	VBUS peak 8.5 V (U201 clamps 0.84 A); VBUS_DET 4.74 V	BQ25798 30 V; TCA9535 6 V	OK (option: RC damper → 6.2 V)
S7	I2C rise time (2.2 kΩ, 111 pF estimated from the layout)	207 ns; 310 ns at +50 % C	≤ 300 ns (400 kHz)	OK, measure at bring-up
S8	Display SCLK 40 MHz over the head cable	3.79 V / -0.49 V (drive 2); 3.30 / 0.00 V (drive 1)	±0.3 V beyond the rails	Firmware: drive strength 1
A	Static stability (CAD mass 750 g)	CoG 3.3 mm ahead of the axle at 56 mm height → tips back at 0.58 m/s²	—	**Action**
B	Drive (N20 6 V, Ø43 wheels)	1:100 → 0.34 m/s, stall force 11.5 N (≫ tip limit)	TB6612 1.2 A cont.	OK with acceleration limit
C	Runtime (44 Wh)	idle 10.1 h · conversation 6.9 h · vision + drive 4.9 h · full 4.3 h · standby 11 days	—	—
D	Charging 10 → 80 %	dock 2.5 A: 1.5 h · dock 1.8 A: 2.2 h · USB-C 5 V 3 A: 4.0 h (Pi idle)	—	USB-C is a backup input
E	Pogo-pin current sharing	at 2.5 A: P(one pin > 1.5 A) = 76 %; at 1.8 A: 0 %	1.5 A per pin (datasheet)	**Action: IINDPM ≤ 1.8 A**
F	Battery protector / fuse	worst load 6.0 A, realistic 4.2 A vs. trip ≥ 6.4 A (12 ms)	fuse 10 A	OK
G	Thermal (Pi 5 SoC inside the shell, 25 °C room)	conversation 60 °C · full load 82 °C	soft throttling 80 °C	Marginal at sustained full load
H	Docking first-try success	as written in the design: 46 % · recommended procedure: 99 %	capture ±6.5 mm	**Action: procedure**
I	AprilTag (40 mm) range, 1536 px stream	detection ≈ 2.0 m, good pose ≤ 0.74 m, closest 0.10 m	—	OK

Findings that need a decision

The design files were not changed, except the pogo-pin text in the dock BOM (finding 2).

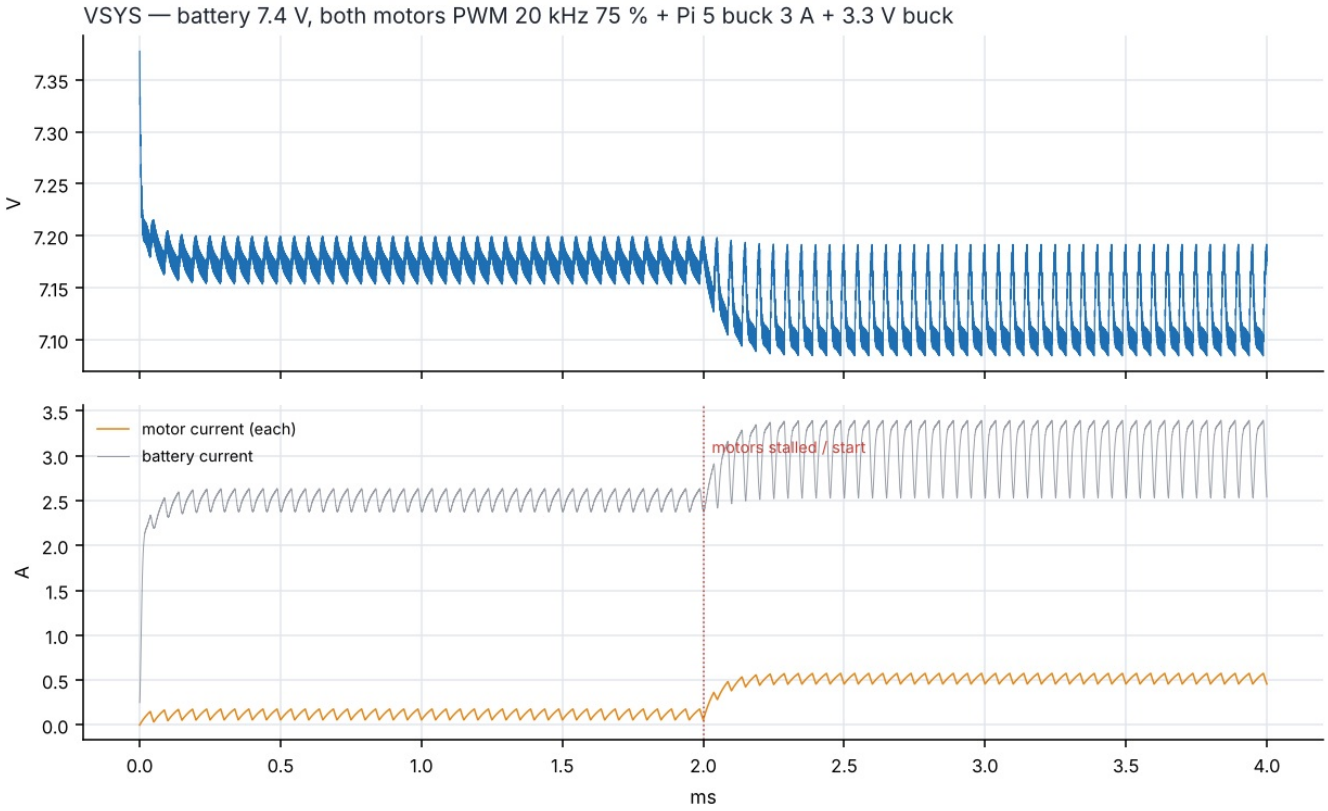
1. **Tip-back margin is small (A).** The centre of gravity is only 3.3 mm in front of the wheel axle (5–95 % range with the mass estimates: 1.7–4.9 mm) and 56 mm high. The caster carries ≈ 50 g. Any acceleration above **0.58 m/s²** forwards (or braking while reversing onto the dock) lifts the caster and rocks the robot back 6.6° onto the rear skirt; a 0.6 N push on the rear of the top does the same. Options: (a) firmware only — limit acceleration and braking to ≤ 0.4 m/s² with a jerk-limited ramp (enough for a desk robot: 0.25 m/s in 0.6 s); (b) **100 g of ballast** under the front of the main board (next to the caster) → CoG 9.8 mm, limit 1.87 m/s²; (c) two small rear glides on the skirt at |X| ≈ 55 mm (outside the dock base), 1–1.5 mm above the floor, which cap the rock at ≈ 1.4°. Recommended: (a) always, plus (b) or (c). Check at the first assembly by weighing the caster load (≈ 50 g expected).
2. **Pogo pins (E).** The YTC1P-8020 datasheet gives 1.5 A max and ≤ 100 mΩ per pin. With two pins per polarity and 2.5 A, unequal contact resistance puts more than 1.5 A on one pin in 76 % of the cases (worst 2.08 A). **Set the BQ25798 VAC2 input current limit to 1.8 A** (worst 1.50 A per pin): dock charging 10 → 80 % takes 2.2 h with the Pi idle (1.7 h with the Pi off) instead of 1.5 h. A hardware alternative for a later revision: three pins per polarity or 3 A pins. The BOM description of the pogo pins was corrected to 1.5 A / 100 mΩ.
3. **Docking procedure (H).** “See the tag, turn 180°, reverse” from 0.40 m succeeds at the first try in 46 % of the runs (σ 11 mm at the opening): the heading error is multiplied by the reverse distance. Recommended procedure: track the tag until the camera is 0.12 m from it, drive on to a turn point 134 mm from the dock wall (the closest point where the shell clears the tongue while turning), turn 180° with the gyro, reverse 62 mm until VAC2 is detected. First try 99 %; without gyro scale and camera-yaw calibration 82 %. Retry automatically when VAC2 is not detected (three tries: > 99.9 %). A larger tongue chamfer (8 mm) alone only brings the 0.40 m procedure to 65 %.

4. **Heat at sustained full load (G).** With ≈ 7.5 W on the Pi 5 (all cores busy, camera on) the SoC reaches ≈ 82 °C in a 25 °C room (soft throttling starts at 80 °C) because the air inside the shell sits ≈ 16 K above the room. Conversation with a cloud AI (≈ 4.5 W) stays at ≈ 60 °C. Doubling the gills and adding a low inlet gains only ≈ 5 K, since most of the rise is the Active Cooler itself. Practical rule: heavy AI in the cloud, or accept throttling during long local inference. Docked and charging in a 35 °C room the air inside reaches ≈ 46 °C: the BQ25798 JEITA “warm” range will reduce the charge (NTC on TS).
5. **Pi 5 rail at extreme load (S2).** At a realistic 1 → 3 A step the Pi sees ≥ 4.90 V. Only a 0.5 → 5 A step at 2 A/µs with a nearly empty battery touches the 4.63 V warning level (4.62 V for ≈ 10 µs, mostly the USB-C cable's inductance). Optional: R301 182 k → 187 k and R302 24 k → 24.3 k sets 5.22 V (4.68 V minimum, 4.95 V steady at 5 A, overshoot 5.31 V < 5.5 V).
6. **USB-C charging (D).** The BQ25798 reads only D+/D− (BC1.2), not the Type-C CC advertisement; a DCP sets 3.25 A with VINDPM protecting the source. From a 5 V 3 A charger the robot charges in 5.4 h with the Pi idle; from a 1.5 A source only 0.31 A reach the battery. The dock is the main charging path; firmware: VINDPM ≈ 4.5 V and ICO enabled for USB-C.
7. **Firmware settings that come out of the simulations:** ESP32-S3 drive strength 1 on GPIO11/12 (S8); acceleration limit (A/B); VAC2 IINDPM 1.8 A (E); motor PWM limited to a 6 V average (B); docking procedure and retries (H).

2. Circuit simulations (ngspice)

S1 — VSYS with the motors under PWM

TB6612 (≈ 0.5 Ω per switch), both N20 channels in phase at 20 kHz / 75 % (worst case), motor 9 Ω / 0.6 mH, start from standstill at 2 ms; Pi 5 buck input as 3 A pulses at 600 kHz; 3.3 V buck 0.25 A. Battery 2S2P ≈ 90 mΩ + 0.3 µH; VSYS bank 72 µF ceramic (derated) + 100 µF electrolytic. Result: 46 mV p-p while driving, 108 mV p-p with both motors stalled, battery current 2.5 A / 3.1 A.

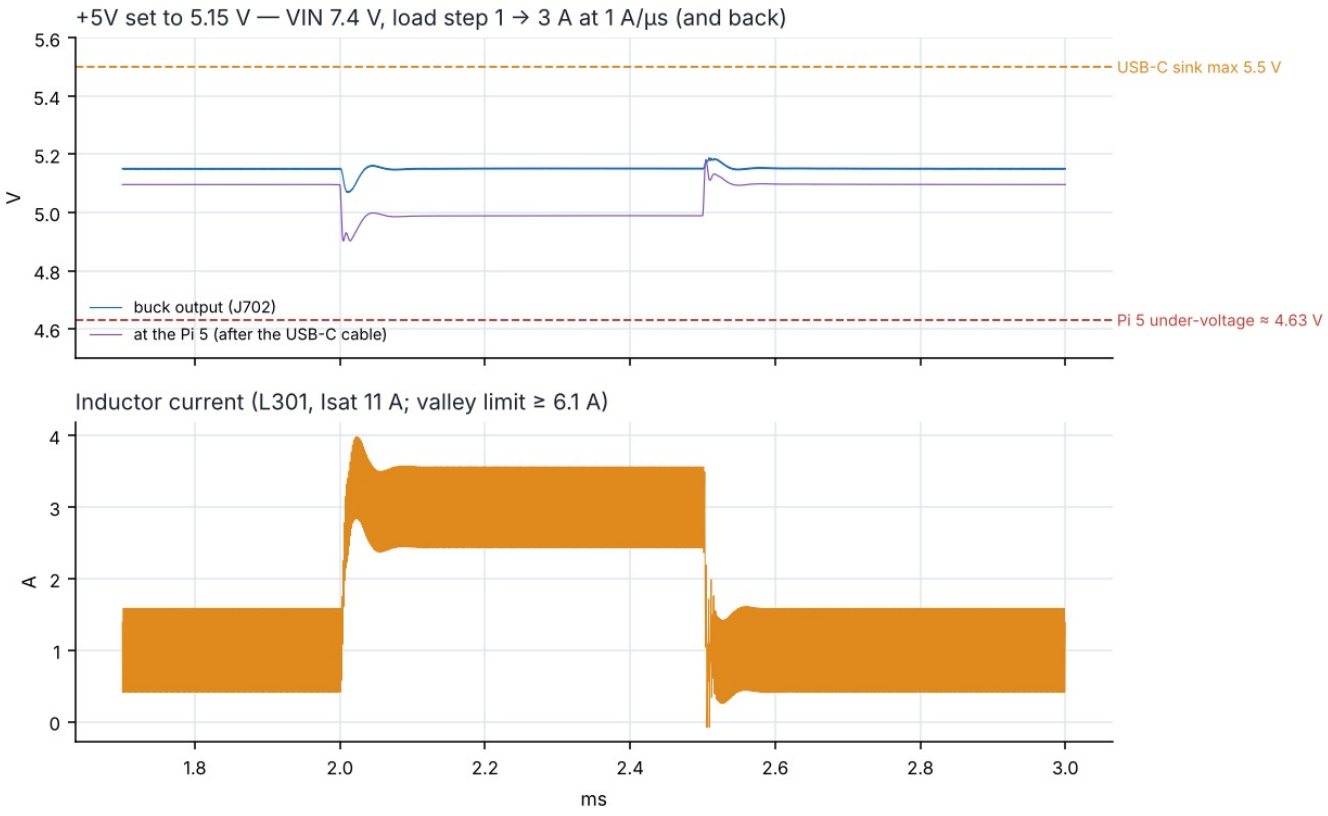


S1 — battery 7.4 V

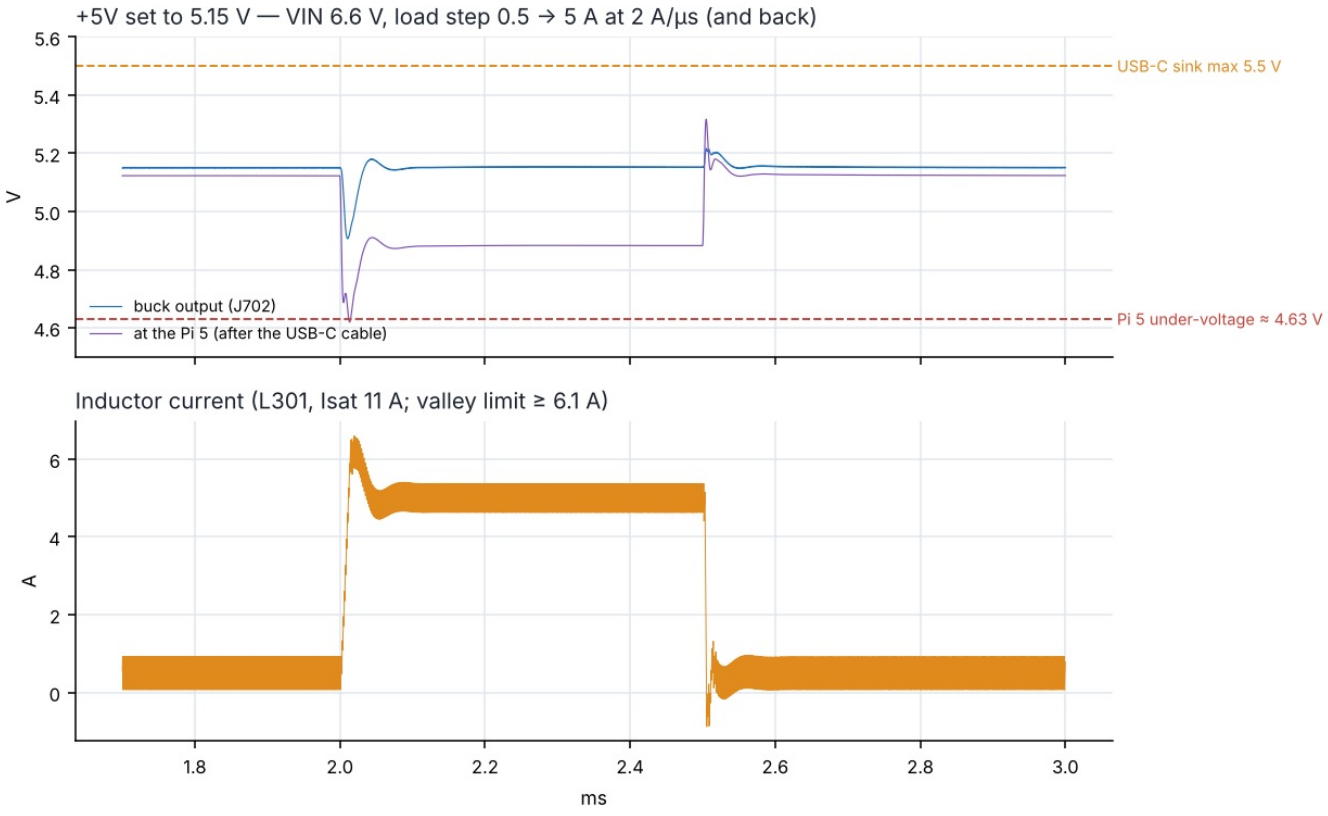
S2 — +5V rail for the Pi 5 (TPS566238, behavioural D-CAP3 model)

Constant-on-time model (on-time Vout/(Vin·fsw), 100 ns minimum off-time, Rds(on) 20.8/10.6 mΩ, VFB 0.6 V), validated at 5150 mV DC, ≈ 600 kHz and 3.9 mV p-p ripple. Load: USB-C cable to the Pi (2 × 25 mΩ, 0.15 µH) and ≈ 20 µF at the Pi input. Cases: a) 7.4 V, 1 → 3 A at 1 A/µs; b) same at 6.6 V; c/d) 0.5 → 5 A at 2 A/µs at 8.4 V / 6.6 V; e) case d with the output set to 5.22 V.

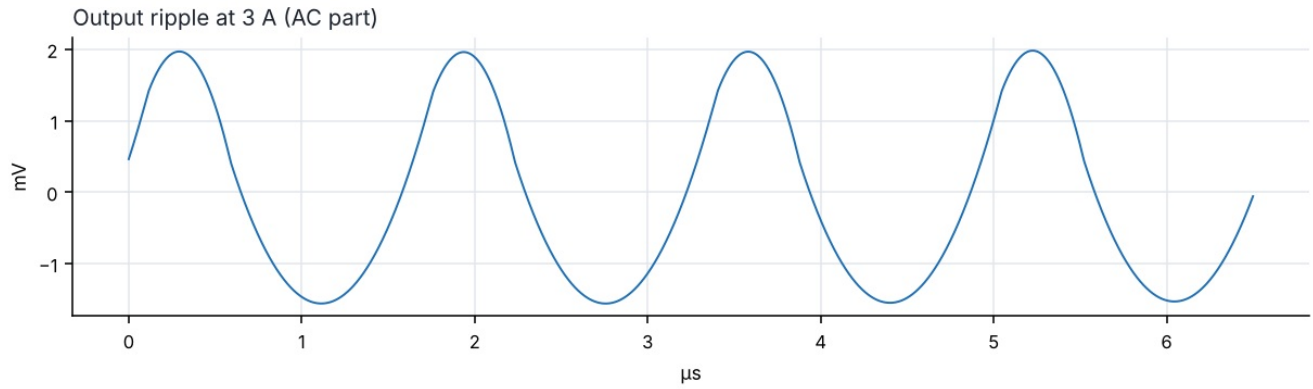
Case	Vout min / max	Pi min	Pi steady (high load)	L301 peak
a	5.07 / 5.19 V	4.90 V	4.99 V	4.0 A
b	5.07 / 5.20 V	4.90 V	4.99 V	3.8 A
c	5.00 / 5.20 V	4.69 V	4.88 V	6.5 A
d	4.91 / 5.22 V	4.62 V	4.88 V	6.6 A
e	4.97 / 5.31 V	4.68 V	4.95 V	6.9 A



S2a — realistic Pi 5 load step



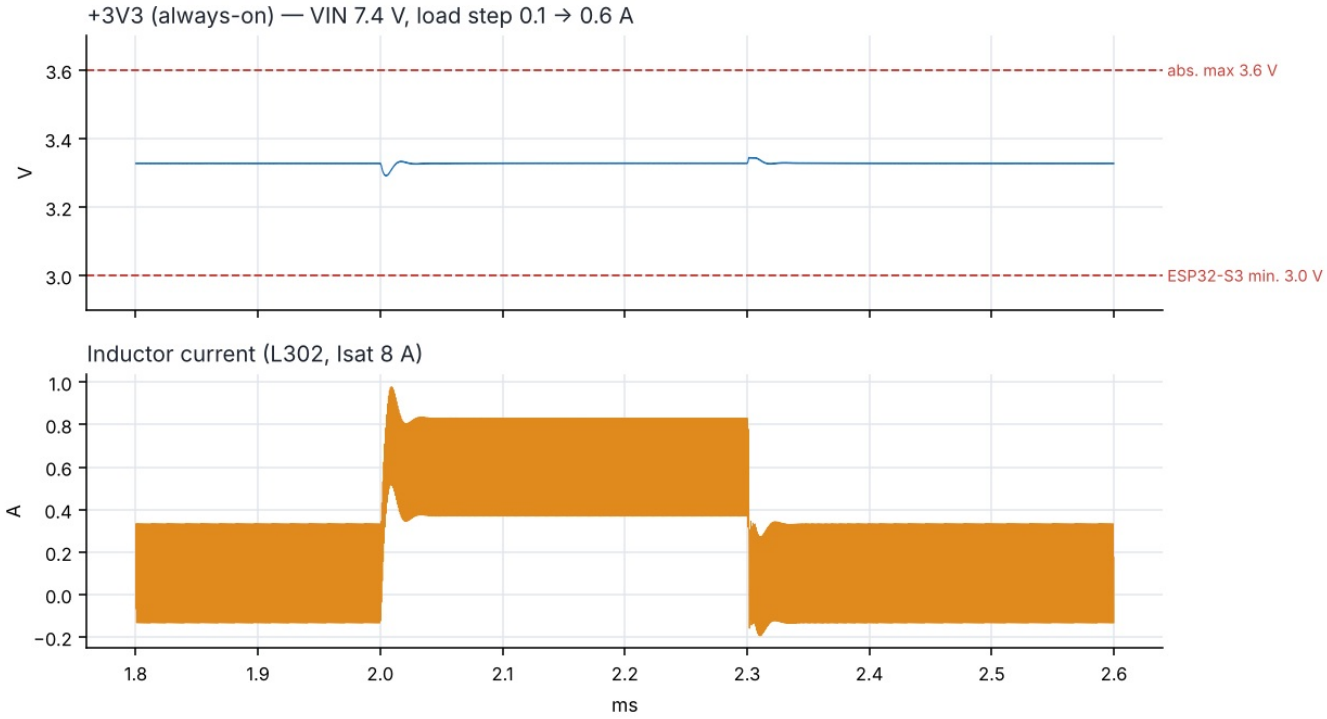
S2d — extreme step at a low battery



S2 — output ripple at 3 A

S3 — +3V3 always-on rail (TPS62933)

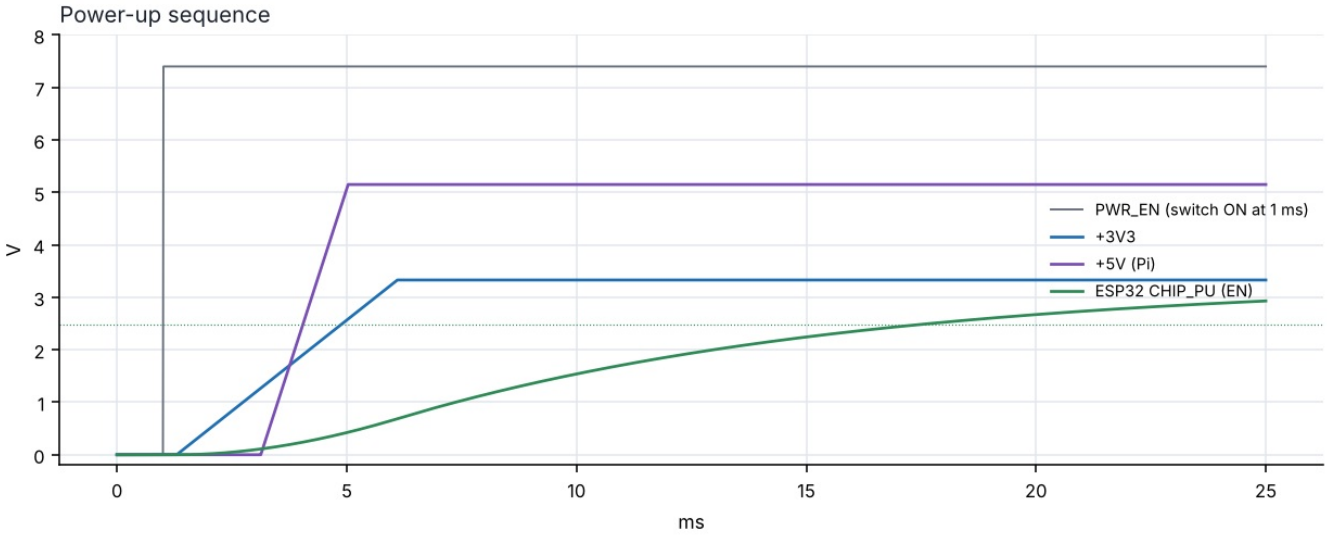
The TPS62933 is peak-current-mode; the COT model at 1.2 MHz gives ripple and the size of the response to a 0.1 → 0.6 A step (Wi-Fi burst + backlight): 3.29–3.35 V, 2.1 mV p-p.



S3 — +3V3 load step

S4 — power-up sequence

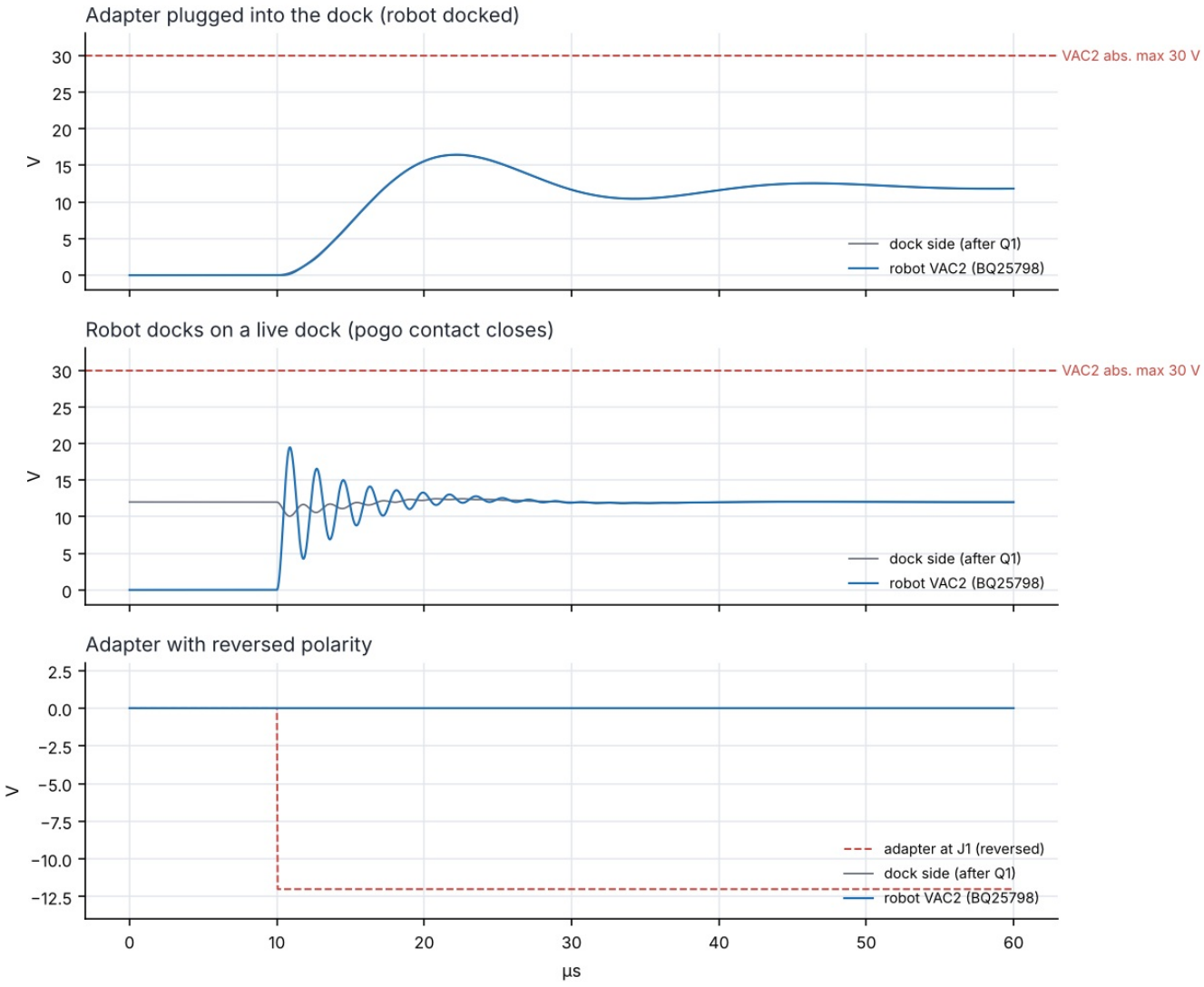
Switch on at 1 ms → TPS62933 soft start (Css 33 nF ≈ 4.8 ms) → PI_EN (R310 100 k from +3V3) crosses the EN threshold at 3.1 ms → +5V soft start 1.9 ms (in regulation at 4.9 ms) → ESP32 CHIP_PU (10 k / 1 µF) reaches 0.75 × VDD at 17.4 ms, ≈ 11 ms after +3V3 is stable.



S4 — sequence

S5 — dock transients

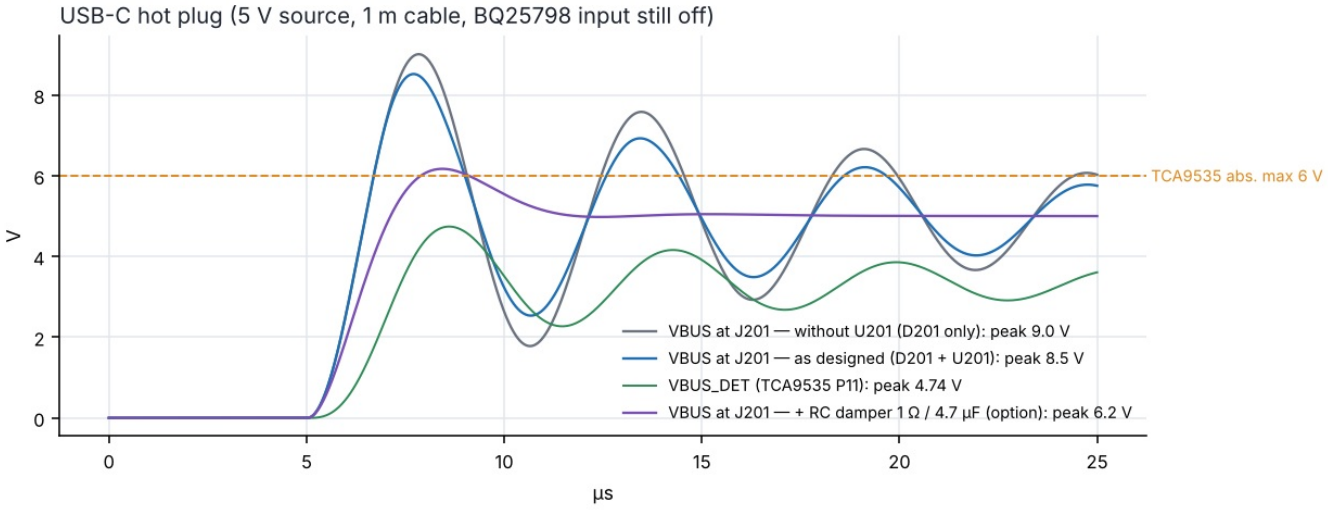
12 V adapter with 1.5 m cable → PTC → AO3401A reverse-polarity FET → 10 µF + SMF15A → pogo pins → robot lead → C202 1 µF + SMF15A → VAC2. Plugging the adapter in gives 16.5 V at VAC2; docking on a live dock rings to 19.5 V (clamped by D202, 2.1 A peak) with a 34 A sub-µs inrush into C202 (12 µC, below the 50 µC that USB allows for a hot plug); a reversed adapter is blocked by Q1.



S5 — dock cases

S6 — USB-C hot plug

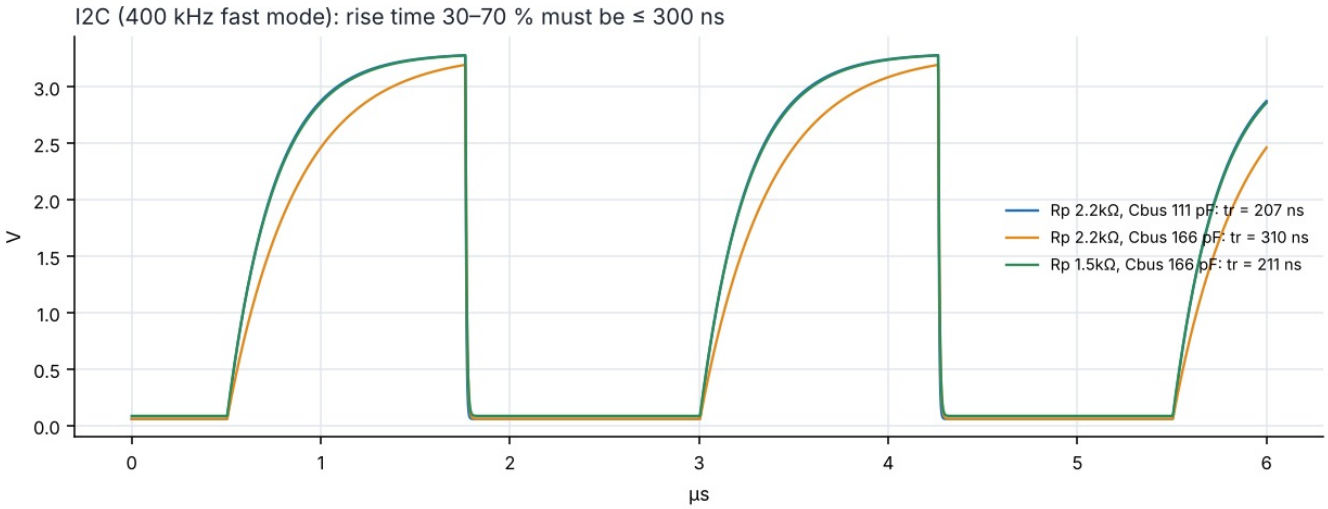
1 m cable (0.12 Ω, 0.8 µH) into C201 1 µF: the LC rings to 9.0 V; U201's VBUS zener (≥ 6 V) limits it to 8.5 V and conducts 0.84 A for ≈ 1 µs. VBUS_DET peaks at 4.74 V (TCA9535 abs. max 6 V, inputs 5 V tolerant). Option for a later revision: 1 Ω + 4.7 µF damper at J201 → 6.2 V and no zener current.



S6 — USB-C hot plug

S7 — I2C edges

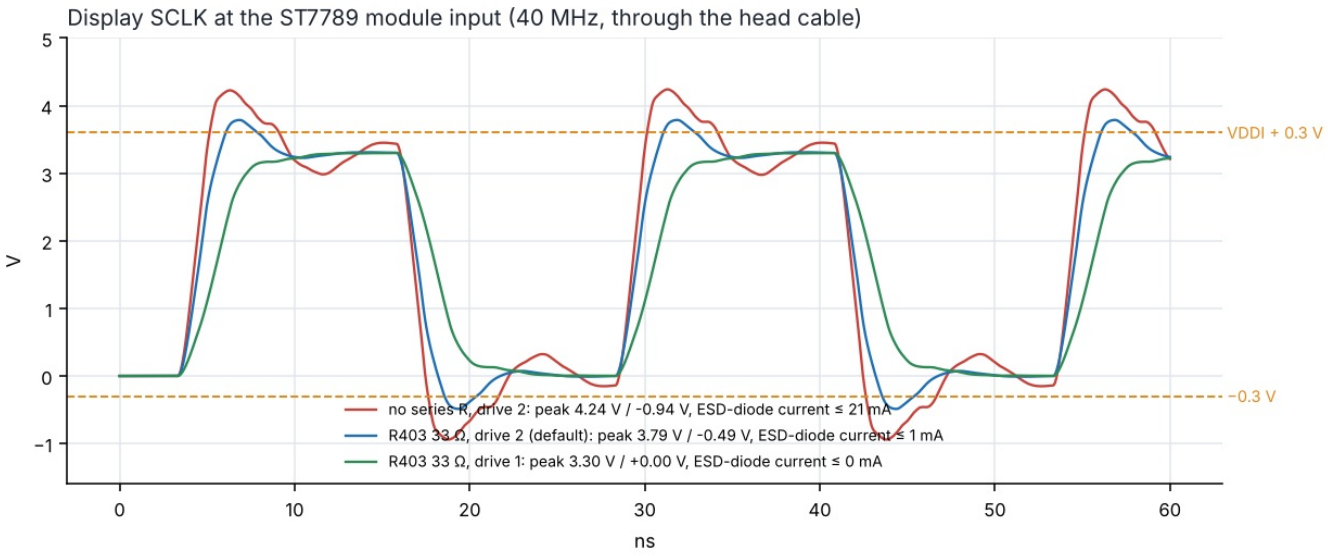
Bus capacitance from the routed boards: 298 mm of trace per line (main 244 mm + head 54 mm), 120 mm of head cable, ≈ 15 vias and 7 devices (ESP32-S3, BQ25798, TCA9535 9.5 pF max, LSM6DSOX, 2 × VL53L4CD, VL53L1X) → 111 pF. 2.2 kΩ: 207 ns (limit 300 ns); with 50 % more capacitance 310 ns; 1.5 kΩ would give 211 ns (3 mA sink limit still met). Measure tr at bring-up.



S7 — SDA rising edges

S8 — display SPI over the head cable

33 Ω series (R403) → 48 mm on the main board → 120 mm cable → 73 mm on the head board → module with ESD diodes. At the default drive strength the edge rings to 3.79 V / -0.49 V (diode current < 1 mA, harmless but outside the ±0.3 V rating); with drive strength 1 the edge is clean (3.30 V / 0.00 V). Without R403 it would reach 4.24 V.

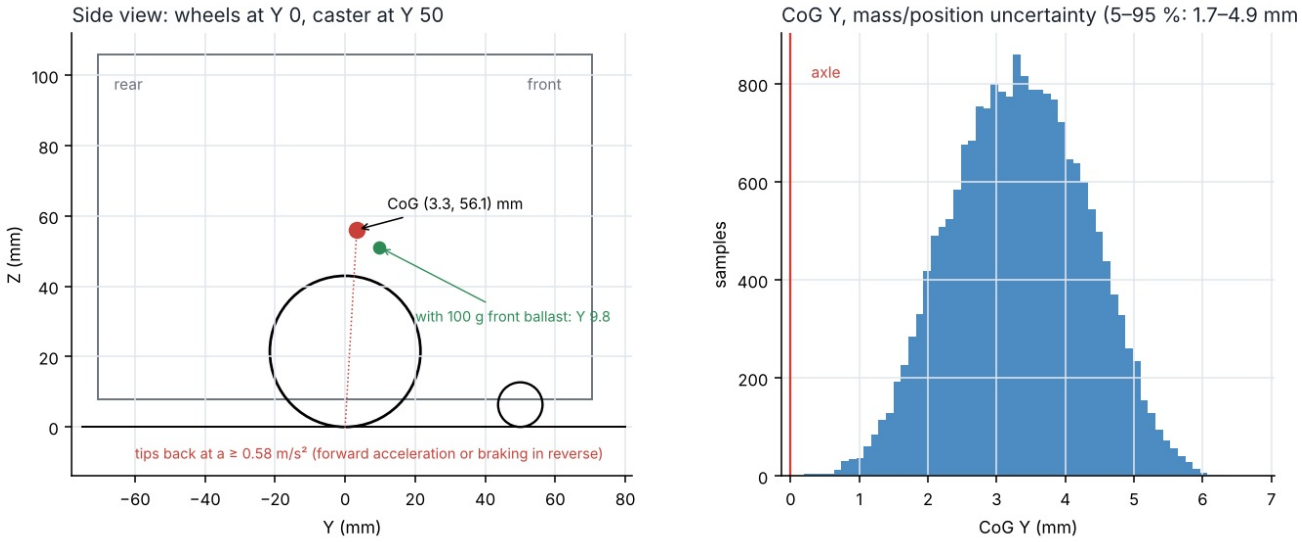


S8 — SCLK at the display

3. System analyses (Python)

A — mass properties and stability

Printed parts from the CAD solids (PETG 1.27 g/cm³ × solid fraction), bought parts from catalogue masses at their CAD positions: **750 g**, CoG (0.5, 3.3, 56.1) mm. Lateral tipping in a turn needs 6.2 m/s² (not a concern); tipping back needs only 0.58 m/s² (see finding 1).



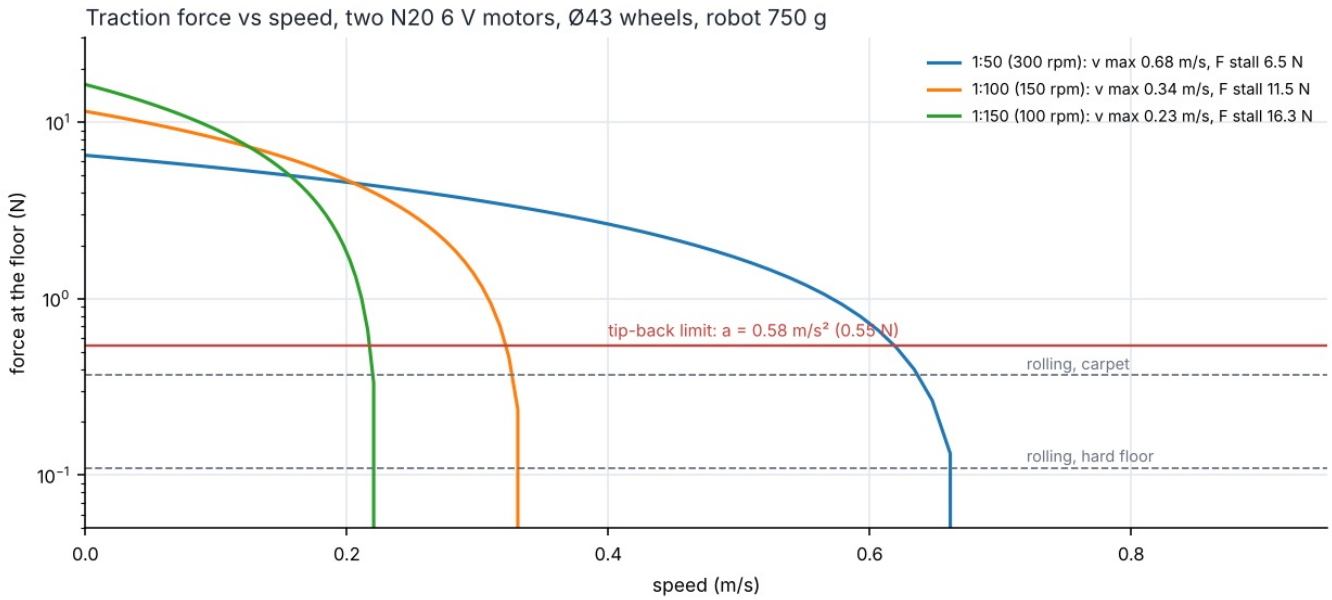
A — CoG and its uncertainty

B — drive

Generic GA12-N20 6 V motor (9 Ω, 15 000 rpm base, est.), gearbox efficiency 0.48–0.65:

Gear	Wheel rpm (6 V)	Max speed	Stall force (2 motors)	Encoder counts / mm
1:30	500	1.13 m/s	4.1 N	6.2
1:50	300	0.68 m/s	6.5 N	10.4
1:100	150	0.34 m/s	11.5 N	20.7
1:150	100	0.23 m/s	16.3 N	31.1
1:298	50	0.11 m/s	29.9 N	61.8

Recommended **1:100 (150 rpm at 6 V)**: 0.34 m/s top speed, fine control down to ≈ 3 cm/s for docking, 20.7 counts/mm. A 0.4 m/s² ramp needs only 0.06 A per motor; stall at 8.4 V is 0.84 A (TB6612 1.2 A continuous). Every option can exceed the tip-back limit, so acceleration must be limited in firmware; keep the motor PWM at a 6 V average (duty ≤ 6 V / VSYS).

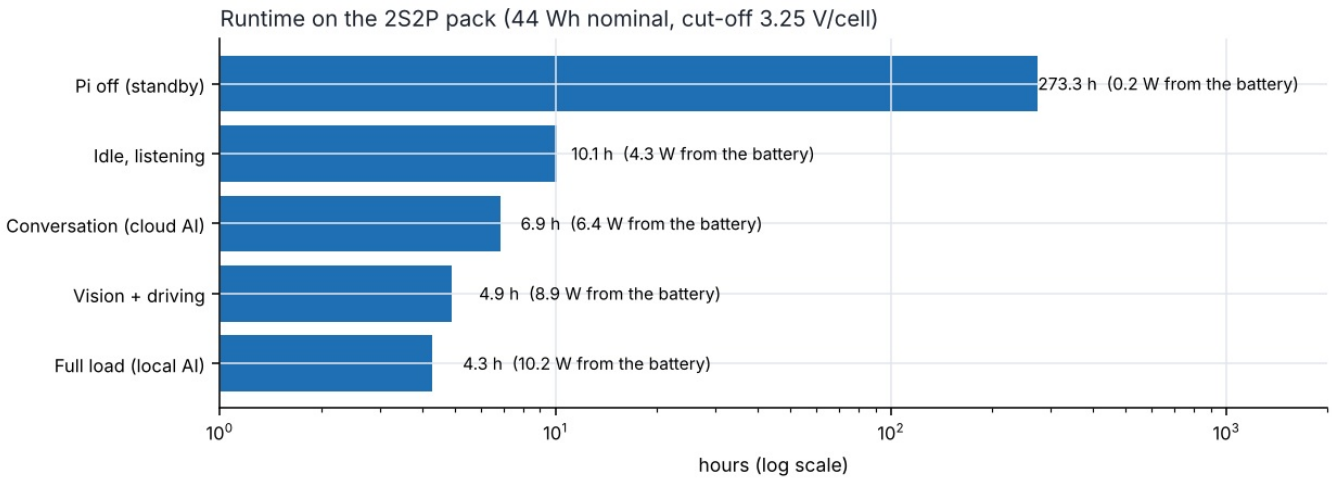


B — force vs speed

C — power budget and runtime

Loads (est., Pi 5 figures from published measurements; camera and XVF3800 powered from the Pi): +5V buck 90–93 %, +3V3 86 %, pack resistance 67 mΩ, cut-off 3.25 V/cell under load.

Mode	Battery power	Runtime
Pi off (standby)	0.2 W	273.3 h
Idle, listening	4.3 W	10.1 h
Conversation (cloud AI)	6.4 W	6.9 h
Vision + driving	8.9 W	4.9 h
Full load (local AI)	10.2 W	4.3 h

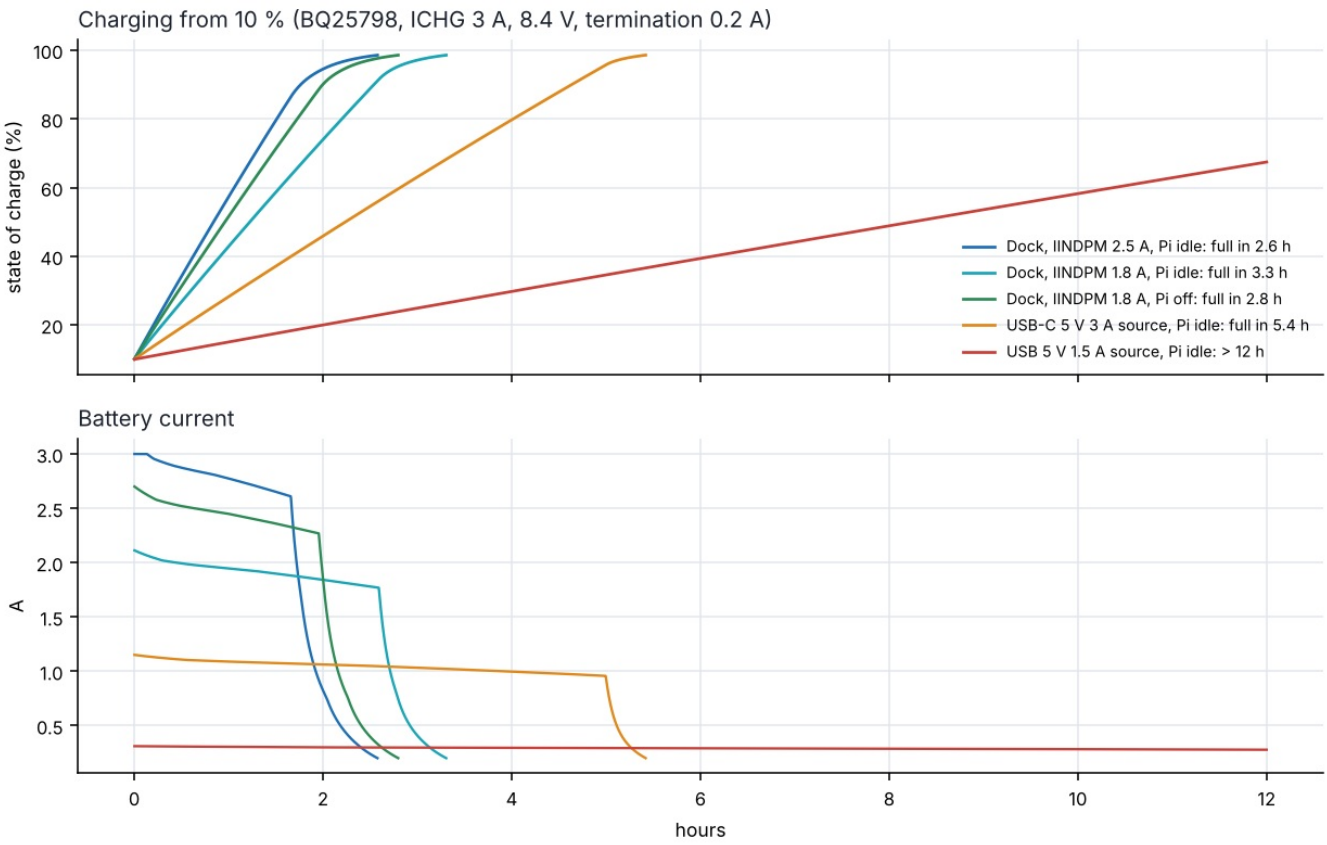


C — runtime

D — charging

BQ25798 with ICHG 3 A, 8.4 V, termination 0.2 A; dock path 12 V − 0.32 Ω × I (cable, PTC, Q1, pogo, input FETs), buck 94 %; USB-C 5 V − 0.14 Ω × I, boost 91 %; battery model with OCV curve, pack resistance and a polarisation branch.

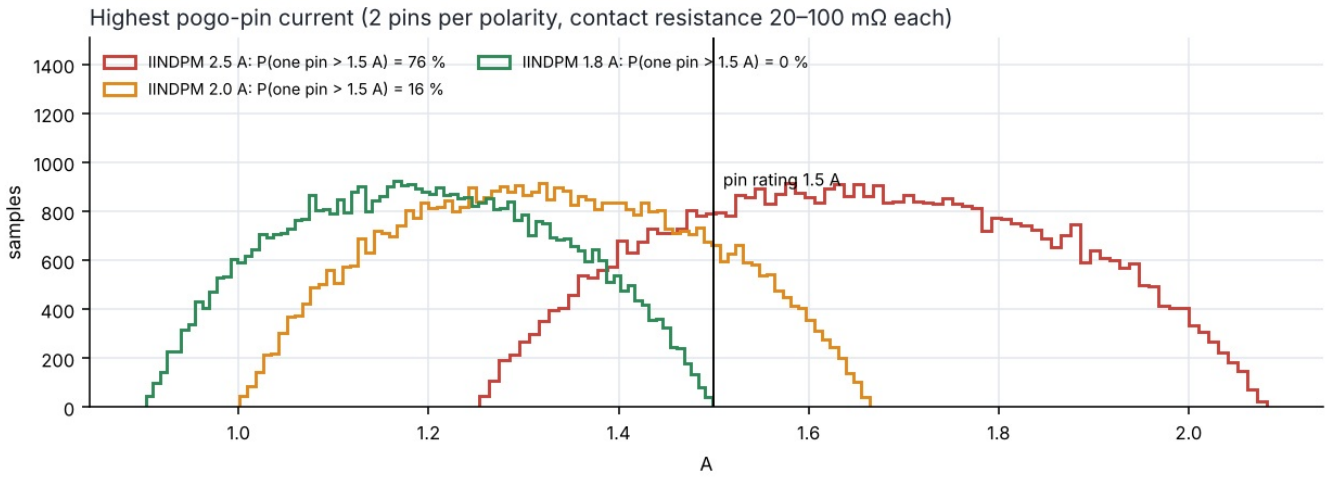
Case	Input power	10 → 80 %	Full
Dock, IINDPM 2.5 A, Pi idle	28.0 W	1.5 h	2.6 h
Dock, IINDPM 1.8 A, Pi idle	20.6 W	2.2 h	3.3 h
Dock, IINDPM 1.8 A, Pi off	20.6 W	1.7 h	2.8 h
USB-C 5 V 3 A source, Pi idle	13.7 W	4.0 h	5.4 h
USB 5 V 1.5 A source, Pi idle	7.2 W	—	> 12 h



D — charging

E — pogo pins

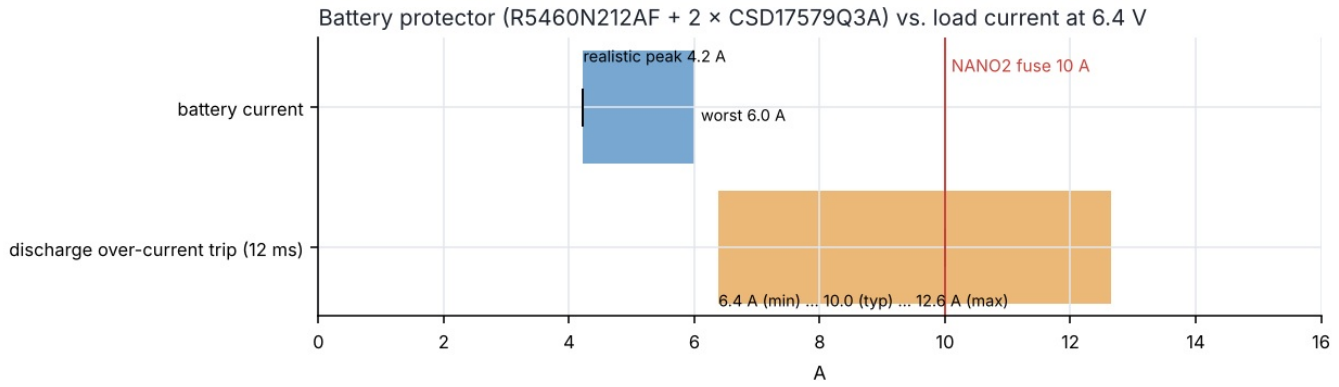
Contact resistance per pin uniformly 20–100 mΩ (datasheet max 100 mΩ), the worse of the two polarity pairs counts. Loop drop at 1.8 A ≤ 0.18 V; ≤ 81 mW per contact.



E — highest pin current

F — battery protector and fuse

R5460N212AF: discharge over-current 0.200 V ± 15 mV (12 ms), short circuit 1.1 V (0.7–1.5 V, 300 µs), charge over-current −0.2 V; 2 × CSD17579Q3A (10.2 mΩ max at 10 V, 14.2 mΩ max at 4.5 V; gate drive ≈ VDD). Trip range 6.4–12.6 A (typ. 10.0 A); short 24–88 A; charge over-current ≥ 5.9 A vs. ICHG 3 A. Over-charge detection 4.29 V/cell vs. 4.20 V charge: the pack has no balancing, so use matched cells from one batch.

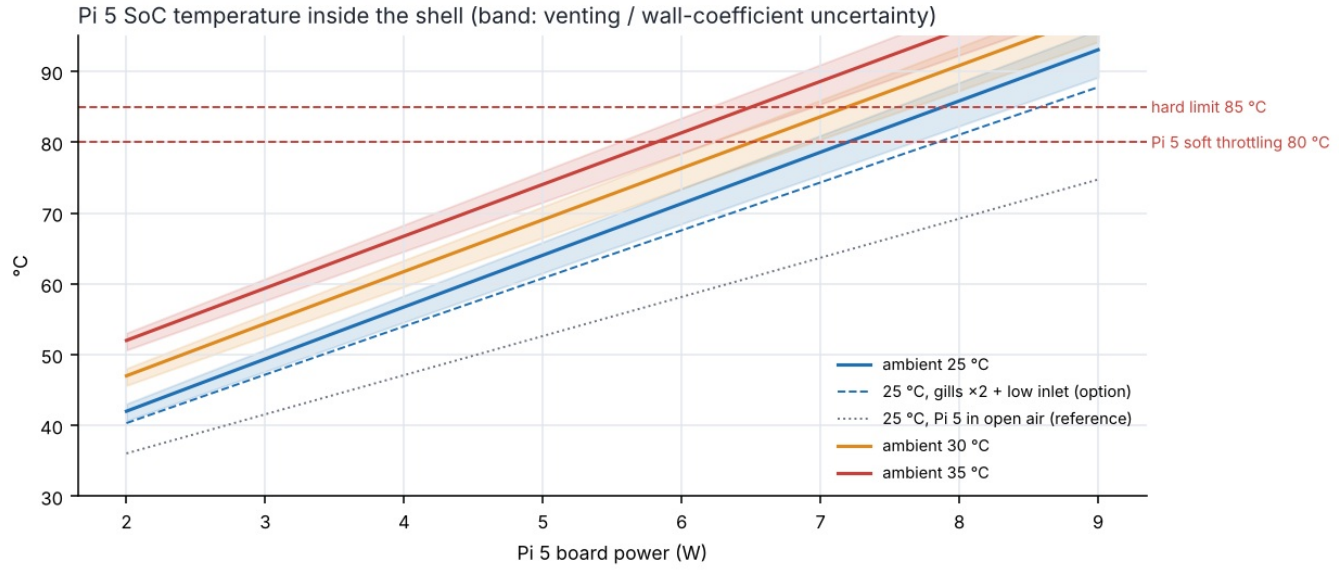


F — protector margin

G — thermal

Lumped model: shell walls ≈ 0.30 W/K, board underside 0.08 W/K, stack-effect flow through the wheel cut-outs and the gills/halo holes, part of the Active Cooler flow exchanged through the gills; SoC ≈ 8.5 K/W above the cooler intake.

Case	Internal air	SoC	SoC with more venting
idle @ 25 °C	33 °C	49 °C	47 °C
idle @ 35 °C	43 °C	59 °C	57 °C
conversation @ 25 °C	36 °C	60 °C	57 °C
conversation @ 35 °C	46 °C	70 °C	67 °C
full load @ 25 °C	41 °C	82 °C	78 °C
full load @ 35 °C	51 °C	92 °C	88 °C
docked + charging, idle @ 25 °C	36 °C	52 °C	49 °C
docked + charging, idle @ 35 °C	46 °C	62 °C	59 °C
docked + charging, full load @ 25 °C	44 °C	85 °C	80 °C
docked + charging, full load @ 35 °C	54 °C	95 °C	90 °C

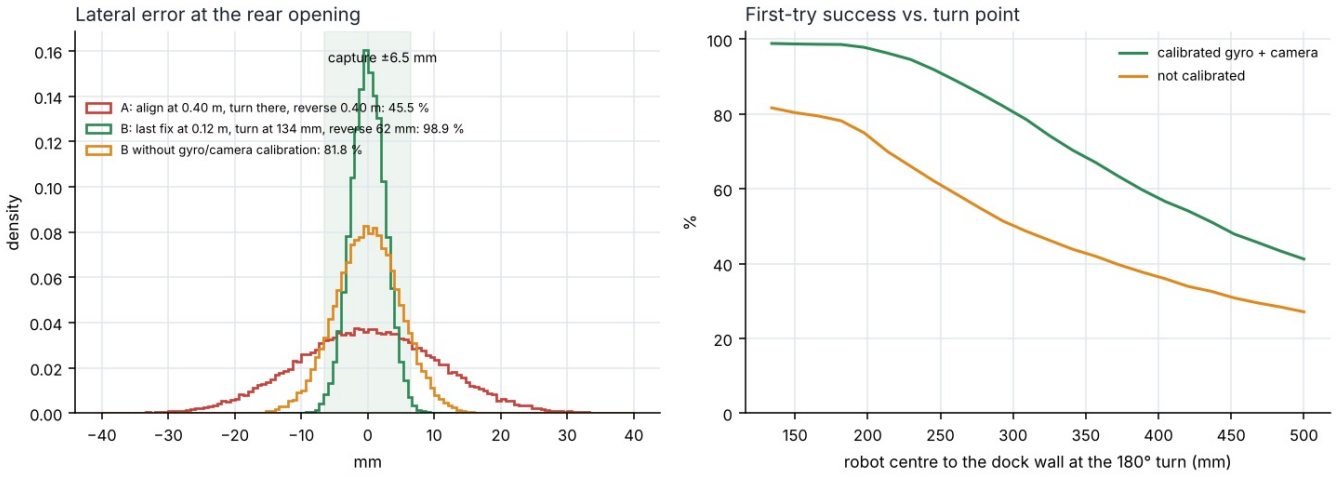


G — Pi 5 SoC temperature

H — docking Monte Carlo (40 000 runs per case)

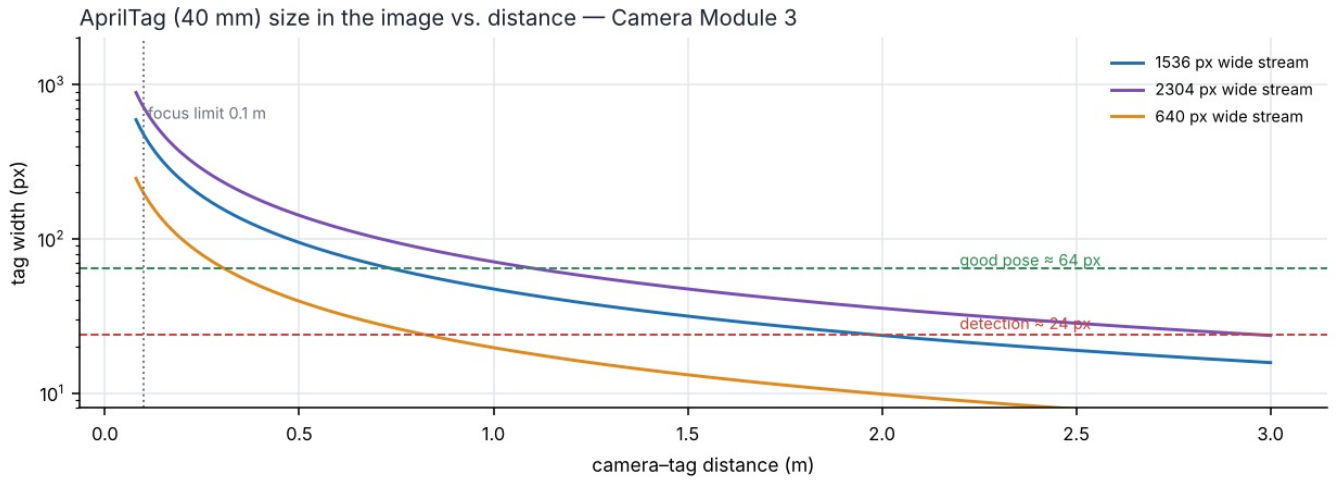
Error sources: tag pose (lateral and yaw, growing with distance), camera-yaw calibration, the blind drive to the turn point, gyro scale error over 180°, turn-centre drift, then the reverse. Capture when the rear opening meets the tongue within ±6.5 mm (2 mm clearance + 4.5 mm chamfer) and ±6°.

Procedure	First try	σ at the opening	Within 3 tries
A: align at 0.40 m, turn there, reverse 0.40 m	46 %	10.8 mm	83.8 %
B: last fix at 0.12 m, turn at 134 mm, reverse 62 mm	99 %	2.6 mm	100.0 %
B without gyro/camera calibration	82 %	4.9 mm	99.4 %
A with an 8 mm tongue chamfer	65 %	10.7 mm	95.6 %



H — docking

I — AprilTag range (Camera Module 3, HFOV 66°)



I — tag size in the image

4. What was not simulated (and how to check it)

Item	Why	Check at bring-up
BQ25798 internal loops, power-path hand-over, ICO	no public model	scope VSYS while plugging/unplugging the dock and USB-C under load
Wi-Fi / BLE (ESP32-S3 antenna in the shell), Pi 5 Wi-Fi	RF/EM tools and the real enclosure needed	RSSI at 1/5/10 m with the shell on and off
EMC / ESD	needs a lab	ESD gun on the dock contacts, USB-C and the touch spot
Acoustics (XVF3800 AEC, speaker in the shell)	room and enclosure dependent	XVF3800 tuning tools, echo-return loss with music
Board-level thermal (TPS566238, BQ25798 hot spots)	no FEA here	thermal camera at 3 A / 5 A and while charging
Printed-part strength, drops	no FEA	0.5 m drop on the skirt corners
Firmware timing (motor loop, PCNT, I2C traffic)	software	logic analyser

5. Files

`sim/spice_sims.py` (S1–S8), `sim/cot.py` (behavioural COT buck), `sim/spice_lib.py` (ngspice runner), `sim/mass_props.py` (CAD mass properties), `sim/system_sims.py` (A–I), `sim/build_sim_report.py` (this report); netlists in `sim/spice/`, results in `sim/out/spice_results.json` and `sim/out/system_results.json`.